# NEXUS

Universal Collectibles Recognition System

## Patent Attorney Package

Prepared for Non-Provisional Conversion

Inventor: Kevin Caravaggio
Rutherford, New Jersey

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**Conversion Deadline: November 27, 2026**

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**CONFIDENTIAL**

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# 1. Executive Summary

NEXUS is a universal collectibles recognition and cataloging system that eliminates the need for physical sorting through a novel zero-sort methodology. Unlike competitors who build complex physical sorting machines, NEXUS assigns sequential call numbers to items regardless of their attributes, using database indexing to enable retrieval by any criteria.

## The Core Innovation

Instead of sorting cards into bins by set, price, or rarity (which requires expensive hardware and limits query flexibility), NEXUS stores cards in sequential order and lets the database do the sorting virtually.

## Key Filing Information

| Item | Details |
|---|---|
| Inventor | Kevin Caravaggio |
| Provisional Filing Date | November 27, 2025 |
| Conversion Deadline | November 27, 2026 |
| Total Claims | 30 (15 independent, 15 dependent) |
| Primary Innovation | Zero-sort call number system |
| Target Market | $300+ billion collectibles industry |
| Development Status | ~39% complete, core functionality operational |

# 2. The Problem NEXUS Solves

## 2.1 Manual Cataloging is Slow

A typical card shop receives bulk collections of 5,000-50,000 cards. Manual entry takes 30-60 seconds per card. At this rate, cataloging 10,000 cards takes 80-160 hours of labor.

## 2.2 Existing Scanners Require Physical Sorting

Competitors like CardMill ($8,000+) and Roca ($15,000+) sort cards into physical bins by attribute. This approach has fundamental limitations:

- Limited query categories (can only sort by what bins exist)
- Expensive mechanical sorting hardware
- Mechanical failure points (jams, misfeeds)
- Does not scale (more categories = more bins = more cost)

## 2.3 No Universal Solution

Current products are game-specific (MTG only, Pokemon only). Shops dealing with multiple games need multiple systems, each with their own learning curve and maintenance requirements.

# 3. The NEXUS Solution

## 3.1 Zero-Sort Methodology (PRIMARY NOVEL CLAIM)

**This is the core innovation and should be the strongest patent claim.**

### How It Works

1. Card is placed on scanner
2. System captures image, runs OCR and recognition
3. Card is identified (name, set, edition, condition indicators)
4. System assigns the NEXT sequential call number (e.g., AA-0001, AA-0002)
5. Card is placed in the current storage box at that position
6. Database stores the mapping: call_number → card_identity → physical_location

### The Key Insight

*The call number has NO relationship to the card's attributes. A $50,000 Black Lotus might be AA-0001, followed by a $0.05 common at AA-0002. The physical location is arbitrary — the database handles all organization.*

### Why This Matters

| Traditional Sorting | NEXUS Zero-Sort |
|---|---|
| Sort by set → 300+ bins needed | No bins — sequential storage |
| Sort by price → can't also sort by set | Query by ANY attribute instantly |
| Mechanical sorter jams | No moving parts in sorting |
| $8,000-$15,000 hardware | $200-$500 hardware |
| Single-game only | Universal (MTG, Pokemon, Yu-Gi-Oh, Sports) |

# 4. System Architecture

## 4.1 Hardware Components

| Component | Specification | Purpose |
|---|---|---|
| OwlSight Cameras (x2) | 64MP stereo pair | 3D surface analysis, foil detection |
| CZUR Camera | Overhead USB | Document-style bulk capture |
| LED Rings (x3) | 16 LEDs each, software-controlled | Consistent, adjustable lighting |
| Robotic Arm | 6-DOF | Card handling (NOT sorting) |
| Processing | Dual Raspberry Pi 5 | Hardware control + Vision/OCR |
| Communication | ZeroMQ | Inter-device messaging |

## 4.2 Software Components

| Component | Function |
|---|---|
| Recognition Engine | Multi-region OCR, set symbol detection, artwork hashing |
| Zero-Sort Library | Call number generation, location tracking, multi-index queries |
| Marketplace (Anielson) | Automatic listing, P2P transactions, 4-8% fees |
| Business Intelligence | Collection valuation, price trends, inventory analytics |

## 4.3 Multi-Game Support

NEXUS is designed to recognize collectibles across multiple categories:

- Currently: Magic: The Gathering (106,000+ cards), Pokemon TCG, Yu-Gi-Oh
- Planned: Sports Cards, Model Trains, Die-Cast, LEGO, Coins, Stamps, Comics

# 5. Patent Claim Overview

The provisional patent contains 30 claims: 15 independent and 15 dependent.

## 5.1 Core Claims (Highest Priority)

| Claim | Description | Novel? | Risk |
|---|---|---|---|
| Claim 7: Zero-Sort | Sequential call numbers independent of attributes | YES | LOW |
| Claim 1: Universal Recognition | Multi-game OCR and identification | YES | MEDIUM |
| Claim 15: Integrated Marketplace | Location-aware listing and fulfillment | YES | LOW |

## 5.2 Claim 7 Detail: Zero-Sort Call Number System

**This is the most important claim and primary competitive moat.**

What It Claims: A method for cataloging physical collectibles wherein items are assigned sequential call numbers independent of item attributes, eliminating physical sorting through database-indexed location tracking.

Why It's Novel: No competitor does this. It is the opposite approach to CardMill/Roca (physical sorting). The call number has NO relationship to card value, set, or rarity.

Prior Art Risk: LOW. Amazon does chaotic storage for warehouses, but not for collectibles with OCR identification and sequential call numbers.

## 5.3 Additional Claims Summary

| Claim Area | Claims | Description |
|---|---|---|
| Hardware | 2, 3, 18, 19, 30 | Scanning chamber, stereo vision, LED specs, robotic arm |
| Software | 4, 5, 6, 11, 12 | OCR, artwork hashing, API integration, ML, condition |
| Business | 8, 9, 10, 15 | B2B+B2C platform, auto-listing, lower fees, marketplace |
| Dependent | 13-14, 16-17, 20-29 | Game-specific OCR, location tracking, variations |

# 6. Competitive Analysis & Prior Art

## 6.1 Abandoned Prior Art — Key Reference

**US20040002387A1 (ABANDONED) — 'Card reader and scanner device and methods of using same'**

Inventor: Daniel Grady | Filed: June 26, 2002 | Status: ABANDONED

This is the closest prior art to NEXUS. It describes OCR-based card scanning with digital conversion for gameplay purposes. However, it lacks all of NEXUS's key innovations:

- NO zero-sort call number system
- NO physical location tracking
- NO multi-game universal recognition
- NO marketplace integration
- NO robotic arm handling
- Purpose was gameplay, NOT inventory/cataloging

*The abandoned status means no licensing required. Attorney should reference this patent to distinguish NEXUS claims by emphasizing zero-sort, location tracking, and marketplace features.*

Link: https://patents.google.com/patent/US20040002387A1/en

**US20230325392A1 (ACTIVE - 2023) — TAG Grading**

Assignee: TAG P, LLC | Filed: February 7, 2023 | Status: ACTIVE

This is TAG Grading's 2023 continuation covering OCR for card identification and automated grading. Part of a patent family with 134+ claims.

**CRITICAL DISTINCTION:**

- TAG's patents focus on GRADING (condition assessment, defect detection, scoring)
- NEXUS focuses on CATALOGING (identification, location tracking, inventory, marketplace)
- TAG answers: 'What grade is this card?'
- NEXUS answers: 'What card is this, where is it, and how do I sell it?'

**NEXUS differentiators NOT in TAG patents:**

- Zero-sort call number system — NOT in TAG

- Physical location tracking — NOT in TAG
- Marketplace integration — NOT in TAG
- B2B shop deployment — NOT in TAG

*If NEXUS adds grading features, may need to license or design around TAG patents.*

Link: https://patents.google.com/patent/US20230325392A1/en

## 6.2 Direct Competitors

| Competitor | Approach | Price | NEXUS Advantage |
|---|---|---|---|
| CardMill | Physical bin sorting | $8,000 | Zero-sort eliminates bins |
| SortSwift | Mechanical sorter | $8,000 | No mechanical failure points |
| Roca | Enterprise scanner | $15,000+ | 10x lower cost, independent |
| TCGPlayer App | Phone camera | Free | Automated bulk processing |
| TAG Grading | OCR for grading | N/A | Cataloging purpose, not grading |

## 6.3 Prior Art Considerations

| Prior Art | Risk | Mitigation Strategy |
|---|---|---|
| Amazon Chaotic Storage | MEDIUM | Emphasize collectibles-specific application with OCR |
| TAG Grading OCR | MEDIUM | Specify cataloging PURPOSE vs grading purpose |
| Library Call Numbers | LOW | Our call numbers are arbitrary, not classification-based |

## 6.4 Claims with NO Prior Art Found

- Zero-sort with sequential call numbers for collectibles
- Database-indexed location tracking replacing physical sorting for trading cards
- Multi-game universal scanner (MTG + Pokemon + Yu-Gi-Oh + Sports in one device)
- B2B + B2C combined platform with shared database
- Call number with no relationship to item attributes

# 7. Market Opportunity

## 7.1 Total Addressable Market: $300+ Billion

| Category | Market Size |
|---|---|
| Trading Cards (MTG, Pokemon, Sports) | $50B+ |
| Sports Memorabilia | $30B+ |
| Coins & Currency | $50B+ |
| Stamps | $10B+ |
| Comics | $15B+ |
| Model Trains, Die-Cast, LEGO, Other | $145B+ |

## 7.2 Business Model

| Revenue Stream | Pricing |
|---|---|
| Hardware Sales/Rental | $200-$500 purchase or $50-$100/month rental |
| SaaS Subscription | $29-$299/month for software access |
| Marketplace Fees | 4-8% transaction fee (vs TCGPlayer 10.25%) |
| Enterprise Licensing | Custom pricing for large operations |

# 8. Current Development Status

| Component | Status | Notes |
|---|---|---|
| Zero-Sort Library | ✓ Working | 26,850 cards cataloged |
| OCR Recognition | ✓ Working | Multi-region extraction |
| Scryfall Integration | ✓ Working | MTG card matching |
| Camera System | ✓ Working | Stereo + overhead operational |
| LED Lighting | ✓ Working | Software-controlled |
| Robotic Arm | In Progress | Hardware present, wiring in progress |
| Marketplace | In Progress | Server infrastructure ready |
| Consumer App | Planned | After B2B launch |

**Overall Completion: Approximately 39%**

# 9. Documents Checklist

The following documents will be provided separately:

## 9.1 Required Documents

- Provisional Patent Filing (83 pages) — E:\MTTGG\LEGAL\PATENTS\
- USPTO Filing Receipt/Confirmation
- System Architecture Diagrams
- API Documentation
- Database Schema

## 9.2 Recommended Documents

- Photos of scanning station hardware
- Demo video of system operation
- Screenshots of software in action
- Competitor research notes
- Business plan / roadmap

# 10. System Architecture

## 10.1 High-Level Architecture

NEXUS V2 operates as a distributed system with three tiers:

- Central Server (Zultan) — Marketplace, GPU OCR, shop synchronization
- Shop Hardware Stack — Brock (OCR/Database) + Snarf (Cameras/Hardware)
- User Interface — Surface Tablet with 10-tab desktop application

## 10.2 Shop Hardware Stack

| Component | Hardware | Function |
|---|---|---|
| Brock (Pi 5) | 8GB RAM, Coral TPU, 160GB HDD | OCR processing, local database |
| Snarf (Pi 5) | 8GB RAM, 3 cameras | Camera control, hardware automation |
| OwlEye Camera | 64MP, CSI connection | High-res card scanning (9152x6944) |
| CZUR Scanner | USB, 4000x3000 | Bulk scanning |
| ESP32 Controllers | 2x USB Serial | LED lighting, sorting arm |
| Surface Tablet | Windows 11 | User interface terminal |

## 10.3 Data Privacy Architecture

**LOCAL ONLY (Never leaves shop):**

- Customer names, emails, phones, addresses
- Payment information, credit card data
- Sales transactions, purchase history

**CLOUD SYNC (Metadata only):**

- Card metadata, inventory counts
- Price data, set information
- Anonymized analytics

*Enforcement: cloud_sync.py sanitizes ALL outbound data with BLOCKED_FIELDS*

# 11. API Documentation

## 11.1 API Servers

| Server | IP:Port | Role |
|---|---|---|
| Brock | 192.168.1.169:5000 | Local shop OCR & database |
| Snarf | 192.168.1.172:5001 | Hardware & camera control |
| Zultan | 192.168.1.152:5000 | Central marketplace |

## 11.2 Key Endpoints

### Brock (Local Shop)

- POST /api/scan — Process card image with Coral TPU OCR
- GET /api/inventory — Get local inventory
- GET /api/customers — Customer list (LOCAL ONLY)
- POST /api/sales/record — Record sale (LOCAL ONLY)

### Snarf (Hardware)

- POST /api/capture — Capture single image
- POST /api/capture/multi — Multi-pass capture (5 exposures, best select)
- POST /api/arm/sort — Activate sorting arm
- POST /api/leds/set — Control LED lighting
- GET /api/video/stream — Live MJPEG stream

### Zultan (Marketplace)

- POST /api/auth/register — User registration
- GET /api/listings — Search marketplace
- POST /api/listings — Create listing (seller auth)
- POST /api/orders — Checkout
- POST /api/ocr/identify — GPU card identification with Scryfall lookup

# 12. Database Schema

## 12.1 Database Locations

| Location | Database | Purpose |
|---|---|---|
| Zultan (Central) | marketplace.db | Users, listings, orders |
| Brock (Shop) | nexus_local.db | Inventory, customers, sales |

## 12.2 Marketplace Tables (Zultan)

- users — User accounts (id, email, password_hash, role)
- sellers — Seller profiles (shop_name, rating, total_sales)
- listings — Card listings (card_name, set_code, condition, price, quantity)
- cart_items — Shopping cart
- orders — Purchase orders (status, subtotal, shipping, tracking)
- order_items — Line items in orders
- reviews — Seller reviews

## 12.3 Local Shop Tables (Brock) — NEVER SYNCS

- inventory — Local card inventory with physical location tracking
- customers — Customer PII (name, email, phone, address) — LOCAL ONLY
- sales — Sales transactions — LOCAL ONLY
- sale_items — Line items — LOCAL ONLY
- scan_history — OCR scan log
- price_cache — Cached Scryfall/TCGPlayer prices

## 12.4 Privacy Enforcement

**BLOCKED_FIELDS in cloud_sync.py:**

customer_name, customer_email, customer_phone, customer_address, customer_id, buyer_name, seller_name, payment_method, credit_card, bank_account, ssn, tax_id

*Tables that NEVER sync: customers, sales, sale_items*

# 13. Questions for Attorney

## 13.1 Priority Questions

**1. Claim Strength: How broad can we make the zero-sort claim (Claim 7) while still being defensible?**

**2. Amazon Prior Art: How do we differentiate from Amazon's chaotic storage for collectibles specifically?**

**3. TAG Grading: Is the cataloging vs grading distinction enough to avoid overlap?**

**4. Timeline & Cost: Realistic timeline and total cost for conversion?**

**5. Continuation Strategy: Should we file a CIP to add claims developed during the year?**

## 13.2 Secondary Questions

- International filing (PCT) — worth it for our market?
- Provisional scope — any obvious gaps?
- Trade secrets — should some aspects remain unpatented?
- Freedom to operate — any patents we might infringe?
- Competitive filing — what if a competitor files before we convert?

## Next Steps

1. Review this package and the attached provisional patent
2. Schedule initial consultation to discuss claim strategy
3. Identify any additional materials needed
4. Provide engagement letter and fee estimate

Budget: $12,000-$20,000 allocated for conversion process

Target Filing: September 2026 (2 months buffer before deadline)

— End of Document —